

**DBAD 4000 – Advanced Database**

**Project Phase 1: Database Modelling**

**DriveEase Car Rental Management Database**

**Technical Report**

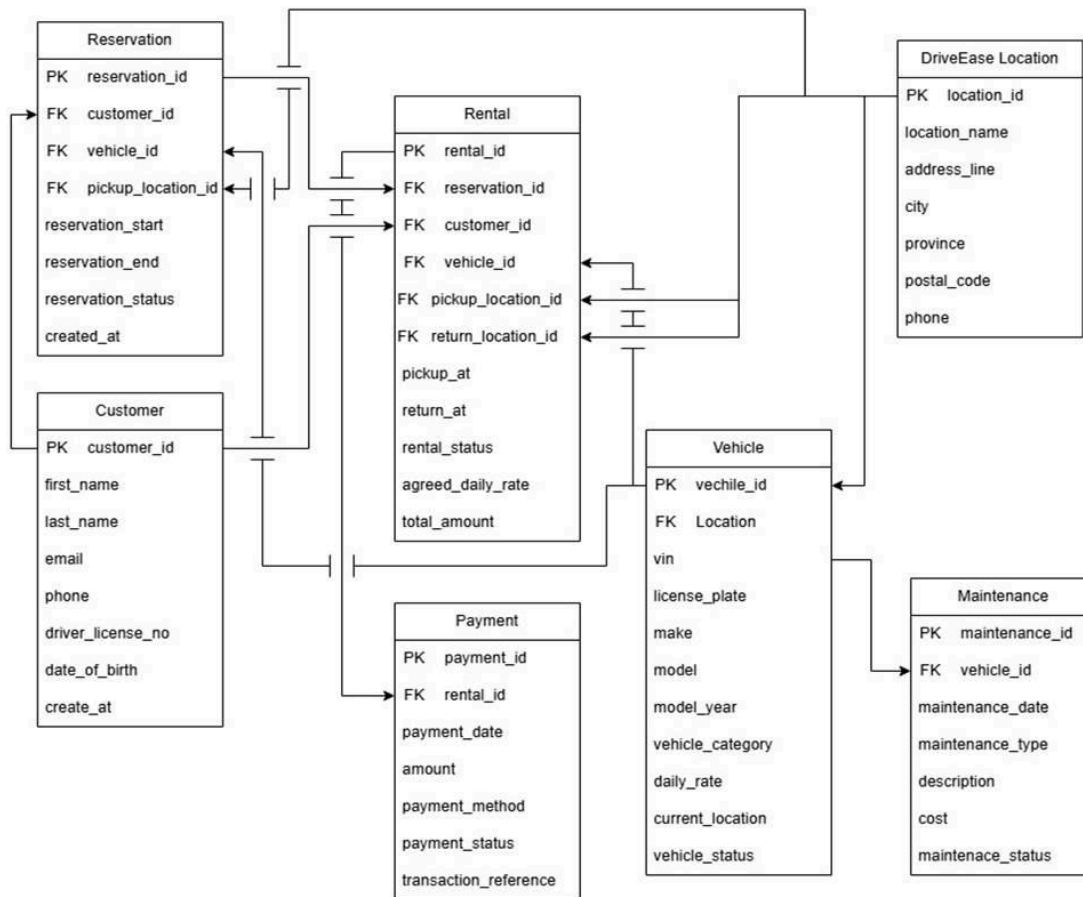
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Course: DBAD 4000 – Advanced Database

Phase: 1 – Database Modelling

## **EER Diagram**

The EER diagram for the DriveEase rental database is shown below. The diagram was created using Draw.io to represent the main entities, their attributes, primary keys, foreign keys, and relationships within the system.



The database consists of seven main entities: CUSTOMER, LOCATION, VEHICLE, RESERVATION, RENTAL, PAYMENT, and MAINTENANCE. The relationships between these entities allow the system to track customers, vehicles, rental locations, reservations, completed rentals, payments, and vehicle maintenance history.

## Relational Schema:

- ® CUSTOMER (customer\_id PK, first\_name, last\_name, email, phone, driver\_license\_no, date\_of\_birth, created\_at).
- ® LOCATION (location\_id PK, location\_name, address\_line, city, province, postal\_code, phone).
- ® VEHICLE (vehicle\_id PK, vin, license\_plate, make, model, model\_year, vehicle\_category, daily\_rate, current\_location\_id FK → LOCATION, vehicle\_status).

- ® RESERVATION (reservation\_id PK, customer\_id FK → CUSTOMER, vehicle\_id FK → VEHICLE, pickup\_location\_id FK → LOCATION, reservation\_start, reservation\_end, reservation\_status, created\_at).
- ® RENTAL (rental\_id PK, reservation\_id FK → RESERVATION, customer\_id FK → CUSTOMER, vehicle\_id FK → VEHICLE, pickup\_location\_id FK → LOCATION, return\_location\_id FK → LOCATION, pickup\_at, return\_at, rental\_status, agreed\_daily\_rate, total\_amount).
- ® PAYMENT (payment\_id PK, rental\_id FK → RENTAL, payment\_date, amount, payment\_method, payment\_status, transaction\_reference).
- ® MAINTENANCE (maintenance\_id PK, vehicle\_id FK → VEHICLE, maintenance\_date, maintenance\_type, description, cost, maintenance\_status).

### Referential Integrity Rules:

| Foreign Key                 | Referenced Key       | Delete / Update Rule                   | Reason                                                        |
|-----------------------------|----------------------|----------------------------------------|---------------------------------------------------------------|
| VEHICLE.current_location_id | LOCATION.location_id | ON DELETE RESTRICT & ON UPDATE CASCADE | A location should not be deleted while vehicles reference it. |
| RESERVATION.customer_id     | CUSTOMER.customer_id | ON DELETE RESTRICT & ON UPDATE CASCADE | Preserves reservation history.                                |
| RESERVATION.vehicle_id      | VEHICLE.vehicle_id   | ON DELETE RESTRICT & ON UPDATE CASCADE | Preserves reservation history.                                |

|                                |                            |                                        |                                              |
|--------------------------------|----------------------------|----------------------------------------|----------------------------------------------|
| RESERVATION.pickup_location_id | LOCATION.location_id       | ON DELETE RESTRICT & ON UPDATE CASCADE | Preserves location history.                  |
| RENTAL.reservation_id          | RESERVATION.reservation_id | ON DELETE RESTRICT & ON UPDATE CASCADE | A rental should retain its reservation link. |
| RENTAL.customer_id             | CUSTOMER.customer_id       | ON DELETE RESTRICT & ON UPDATE CASCADE | Preserves rental history.                    |
| RENTAL.vehicle_id              | VEHICLE.vehicle_id         | ON DELETE RESTRICT & ON UPDATE CASCADE | Preserves vehicle rental history.            |
| RENTAL.pickup_location_id      | LOCATION.location_id       | ON DELETE RESTRICT & ON UPDATE CASCADE | Preserves pickup history.                    |
| RENTAL.return_location_id      | LOCATION.location_id       | ON DELETE RESTRICT & ON UPDATE CASCADE | Preserves return history.                    |

|                        |                    |                                                       |                                                       |
|------------------------|--------------------|-------------------------------------------------------|-------------------------------------------------------|
| PAYMENT.rental_id      | RENTAL.rental_id   | ON<br>DELETE<br>RESTRICT<br>& ON<br>UPDATE<br>CASCADE | Payments<br>must remain<br>linked to their<br>rental. |
| MAINTENANCE.vehicle_id | VEHICLE.vehicle_id | ON<br>DELETE<br>RESTRICT<br>& ON<br>UPDATE<br>CASCADE | Preserves<br>maintenance<br>history.                  |

## Normalization to 3NF:

### 4.3.1 Functional Dependencies:

- ® CUSTOMER: customer\_id → first\_name,last\_name, email, phone, driver\_license\_no,date\_of\_birth,created\_at.
- ® LOCATION: location\_id → location\_name,address\_line, city, province,postal\_code, phone.
- ® VEHICLE: vehicle\_id → vin,license\_plate, make, model,model\_year, vehicle\_category, daily\_rate, current\_location\_id, vehicle\_status.
- ® RESERVATION: reservation\_id → customer\_id, vehicle\_id,pickup\_location\_id, reservation\_start, reservation\_end, reservation\_status, created\_at.
- ® RENTAL: rental\_id → reservation\_id, customer\_id, vehicle\_id, pickup\_location\_id, return\_location\_id,pickup\_at,return\_at,rental\_status,agreed\_daily\_rate,total\_amount.
- ® PAYMENT: payment\_id → rental\_id,payment\_date,amount,payment\_method, payment\_status, transaction\_reference.
- ® MAINTENANCE: maintenance\_id → vehicle\_id,maintenance\_date, maintenance\_type, description, cost, maintenance\_status.

4.3.2 1NF: The design is in First Normal Form because each attribute contains a single atomic value. For example, a customer has a separate first name, last name, email, and phone attributes. Multiple payments are stored as separate Payment rows rather than as repeated payment columns in Rental.

4.3.3 2NF: The design is in Second Normal Form because the tables use single-column primary keys. Therefore, no non-key attribute can depend on only part of a composite primary key. Each non-key attribute depends on the complete primary key of its table.

4.3.4 3NF: The design is in Third Normal Form because non-key attributes describe the entity identified by the primary key and are not stored through unnecessary transitive dependencies. For example, location details are stored in LOCATION rather than repeated in VEHICLE, RESERVATION, and RENTAL. Vehicle details are stored in VEHICLE rather than repeated in RENTAL.

This separation reduces duplication and makes future updates more reliable. It also gives later phases a clean operational source for generating synthetic high-volume data and loading a separate analytical star schema.

## Data Dictionary:

| Table    | Attribute         | Description                  | PostgreS QL Type | Key                    | Validati on                          | Sample / Default |
|----------|-------------------|------------------------------|------------------|------------------------|--------------------------------------|------------------|
| CUSTOMER | customer_id       | Unique customer identifier   | integer          | PK; generated identity | NOT NULL; UNIQUE                     | Generated        |
| CUSTOMER | first_name        | Customer first name          | varchar(50)      | —                      | NOT NULL                             | Alex             |
| CUSTOMER | last_name         | Customer last name           | varchar(50)      | —                      | NOT NULL                             | Smith            |
| CUSTOMER | email             | Customer email address       | varchar(254)     | —                      | NOT NULL; UNIQUE; valid email format | Alex@gmail.com   |
| CUSTOMER | phone             | Customer phone number        | varchar(20)      | —                      | NOT NULL                             | 403-555-0100     |
| CUSTOMER | driver_license_no | Unique driver licence number | varchar(30)      | —                      | NOT NULL; UNIQUE                     | AB123456         |
| CUSTOMER | date_of_birth     | Customer date of birth       | date             | —                      | NOT NULL; reasonable adult/rental    | 1995-06-15       |

|          |               |                                       |              |                        |                             |                   |
|----------|---------------|---------------------------------------|--------------|------------------------|-----------------------------|-------------------|
|          |               |                                       |              |                        | validation                  |                   |
| CUSTOMER | created_at    | Date/time customer record was created | timestampz   | —                      | NOT NULL                    | Current timestamp |
| LOCATION | location_id   | Unique rental location identifier     | integer      | PK; generated identity | NOT NULL; UNIQUE            | Generated         |
| LOCATION | location_name | Branch name                           | varchar(100) | —                      | NOT NULL                    | Downtown Calgary  |
| LOCATION | address_line  | Street address                        | varchar(150) | —                      | NOT NULL                    | 100 Main Street   |
| LOCATION | city          | City                                  | varchar(80)  | —                      | NOT NULL                    | Calgary           |
| LOCATION | province      | Canadian province                     | varchar(50)  | —                      | NOT NULL                    | Alberta           |
| LOCATION | postal_code   | Canadian postal code                  | varchar(7)   | —                      | NOT NULL; format validation | T2P 1A1           |
| LOCATION | phone         | Branch phone number                   | varchar(20)  | —                      | NOT NULL                    | 403-888-2222      |
| VEHICLE  | vehicle_id    | Unique vehicle identifier             | integer      | PK; generated identity | NOT NULL; UNIQUE            | Generated         |
| VEHICLE  | vin           | Vehicle identification number         | varchar(17)  | —                      | NOT NULL; UNIQUE            | 1HGCM826          |
| VEHICLE  | license_plate | Vehicle plate number                  | varchar(15)  | —                      | NOT NULL; UNIQUE            | ABC123            |
| VEHICLE  | make          | Vehicle manufacturer                  | varchar(50)  | —                      | NOT NULL                    | Toyota            |
| VEHICLE  | model         | Vehicle model                         | varchar(50)  | —                      | NOT NULL                    | Corolla           |

|             |                     |                               |               |                        |                                 |                  |
|-------------|---------------------|-------------------------------|---------------|------------------------|---------------------------------|------------------|
| VEHICLE     | model_year          | Vehicle model year            | smallint      | —                      | NOT NULL; reasonable year range | 2025             |
| VEHICLE     | vehicle_category    | Rental category               | varchar(30)   | —                      | NOT NULL; controlled values     | SUV              |
| VEHICLE     | daily_rate          | Standard daily rental rate    | numeric(10,2) | —                      | NOT NULL; >= 0                  | 89.99            |
| VEHICLE     | current_location_id | Vehicle's current branch      | integer       | FK → LOCATION          | NOT NULL                        | 1                |
| VEHICLE     | vehicle_status      | Current vehicle status        | varchar(20)   | —                      | NOT NULL; controlled values     | Available        |
| RESERVATION | reservation_id      | Unique reservation identifier | integer       | PK; generated identity | NOT NULL; UNIQUE                | Generated        |
| RESERVATION | customer_id         | Customer making reservation   | integer       | FK → CUSTOMER          | NOT NULL                        | 1                |
| RESERVATION | vehicle_id          | Vehicle reserved              | integer       | FK → VEHICLE           | NOT NULL                        | 10               |
| RESERVATION | pickup_location_id  | Planned pickup branch         | integer       | FK → LOCATION          | NOT NULL                        | 1                |
| RESERVATION | reservation_start   | Planned rental start          | timestampz    | —                      | NOT NULL                        | 2026-10-10 09:00 |
| RESERVATION | reservation_end     | Planned rental end            | timestampz    | —                      | NOT NULL; after start           | 2026-10-13 09:00 |
| RESERVATION | reservation_status  | Current reservation status    | varchar(20)   | —                      | NOT NULL; controlled values     | Confirmed        |

|             |                    |                                      |               |                        |                                   |                   |
|-------------|--------------------|--------------------------------------|---------------|------------------------|-----------------------------------|-------------------|
| RESERVATION | created_at         | Reservation creation time            | timestamp tz  | —                      | NOT NULL                          | Current timestamp |
| RENTAL      | rental_id          | Unique rental transaction identifier | integer       | PK; generated identity | NOT NULL; UNIQUE                  | Generated         |
| RENTAL      | reservation_id     | Source reservation                   | integer       | FK → RESERVATION       | NOT NULL                          | 25                |
| RENTAL      | customer_id        | Customer renting vehicle             | integer       | FK → CUSTOMER          | NOT NULL                          | 1                 |
| RENTAL      | vehicle_id         | Vehicle rented                       | integer       | FK → VEHICLE           | NOT NULL                          | 10                |
| RENTAL      | pickup_location_id | Actual pickup branch                 | integer       | FK → LOCATION          | NOT NULL                          | 1                 |
| RENTAL      | return_location_id | Actual return branch                 | integer       | FK → LOCATION          | NOT NULL                          | 2                 |
| RENTAL      | pickup_at          | Actual pickup date/time              | timestamp tz  | —                      | NOT NULL                          | 2026-10-10 09:10  |
| RENTAL      | return_at          | Actual return date/time              | timestamp tz  | —                      | NULL until returned; after pickup | 2026-10-13 10:00  |
| RENTAL      | rental_status      | Current rental status                | varchar(20)   | —                      | NOT NULL; controlled values       | Completed         |
| RENTAL      | agreed_daily_rate  | Rate agreed for rental               | numeric(10,2) | —                      | NOT NULL; >= 0                    | 89.99             |
| RENTAL      | total_amount       | Total rental charge                  | numeric(12,2) | —                      | NOT NULL; >= 0                    | 269.97            |
| PAYMENT     | payment_id         | Unique payment identifier            | integer       | PK; generated identity | NOT NULL; UNIQUE                  | Generated         |

|             |                       |                               |               |                        |                             |                  |
|-------------|-----------------------|-------------------------------|---------------|------------------------|-----------------------------|------------------|
| PAYMENT     | rental_id             | Rental paid by transaction    | integer       | FK → RENTAL            | NOT NULL                    | 100              |
| PAYMENT     | payment_date          | Payment date/time             | timestampz    | —                      | NOT NULL                    | 2026-10-13 10:15 |
| PAYMENT     | amount                | Payment amount                | numeric(12,2) | —                      | NOT NULL; > 0               | 269.97           |
| PAYMENT     | payment_method        | Payment method                | varchar(20)   | —                      | NOT NULL; controlled values | Credit Card      |
| PAYMENT     | payment_status        | Payment processing status     | varchar(20)   | —                      | NOT NULL; controlled values | Completed        |
| PAYMENT     | transaction_reference | External/payment reference    | varchar(80)   | —                      | UNIQUE when present         | TXN-10001        |
| MAINTENANCE | maintenance_id        | Unique maintenance identifier | integer       | PK; generated identity | NOT NULL; UNIQUE            | Generated        |
| MAINTENANCE | vehicle_id            | Vehicle receiving maintenance | integer       | FK → VEHICLE           | NOT NULL                    | 10               |
| MAINTENANCE | maintenance_date      | Date maintenance occurred     | date          | —                      | NOT NULL                    | 2026-09-20       |
| MAINTENANCE | maintenance_type      | Type of maintenance           | varchar(50)   | —                      | NOT NULL                    | Oil Change       |
| MAINTENANCE | description           | Maintenance details           | varchar(500)  | —                      | NULL allowed                | Routine service  |
| MAINTENANCE | cost                  | Maintenance cost              | numeric(10,2) | —                      | NOT NULL; >= 0              | 125.50           |
| MAINTENANCE | maintenance_status    | Maintenance status            | varchar(20)   | —                      | NOT NULL; controlled values | Completed        |

## Major Design Decisions

The DriveEase database was designed to focus on the core vehicle rental business using synthetic data. Vehicles have unique VINs and license plates and are assigned to one current location. Reservations and actual rentals are stored separately, while pickup and return locations are tracked independently. Payments and maintenance are also separated to support multiple transactions and maintenance records per rental or vehicle. The design also preserves historical records for reporting and auditing, while vehicle availability and overlapping rental rules will be handled through business rules in later phases.

### Individual Contribution Table

| Team member                  | Responsibilities                                          | Specific Contribution to this Phase                                       |
|------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------|
| <b><i>Vivek Patel</i></b>    | Created the initial relational schema and data dictionary | Worked on relational schema and data dictionary. Made many design choices |
| <b><i>Saihaj Mann</i></b>    | Create an EER diagram.                                    | Created EER diagram.                                                      |
| <b>Kyle Ince</b>             | Technical Sheet; EER diagram review;                      | Did the write up of the technical sheet; reviewed the EER diagram         |
| <b><i>Nathan Woyessa</i></b> |                                                           |                                                                           |

